

Setu WX1 BLE to Wi-Fi IoT Gateway Datasheet

Product Overview

Setu WX1 is a compact and industrial-grade **BLE to Wi-Fi IoT gateway**, designed to bridge Bluetooth Low Energy (BLE) devices with local servers and cloud platforms over standard Wi-Fi networks. It enables seamless data aggregation, protocol translation, and secure IP-based communication without requiring internet connectivity on the end BLE devices.

The gateway continuously scans, connects, and collects data from multiple BLE sensors, beacons, meters, and edge devices, and reliably forwards this data to on-premise systems or cloud platforms using **MQTT, HTTP/HTTPS, and TCP/IP(Optional)** protocols. This makes Setu WX1 ideal for environments where low-power BLE devices need centralized connectivity and management.

Built for continuous indoor operation, Setu WX1 features a low-power, fanless design, stable power architecture, and an industrial enclosure suitable for commercial, healthcare, retail, and industrial deployments. Local buffering and store-and-forward mechanisms ensure data reliability even during temporary network outages.

Whether used for asset tracking, occupancy monitoring, environmental sensing, healthcare device integration, or smart building analytics, **Setu WX1 delivers a secure, scalable, and gateway-centric BLE to Wi-Fi connectivity solution** for modern IoT deployments.





Key Features

BLE to Wi-Fi Protocol Gateway

Seamlessly bridges Bluetooth Low Energy (BLE) devices to IP-based networks over standard Wi-Fi connectivity.

Standard IP Protocol Support

Supports MQTT, HTTP, HTTPS, and TCP/IP for easy integration with local servers, SCADA systems, and cloud platforms.

Secure Wi-Fi Communication

Encrypted Wi-Fi backhaul with secure authentication and credential-based access for protected data transmission.

Low-Power, Fanless Design

Optimized for continuous indoor operation with silent, low-heat performance in commercial and industrial environments.

Gateway-Centric Architecture

Eliminates the need for internet connectivity on BLE end devices, simplifying large-scale deployments.

Multi-Device BLE Aggregation

Concurrently scans, connects, and manages data from multiple BLE sensors, beacons, and edge devices.

Local Buffering & Store-and-Forward

Ensures data reliability by buffering BLE data during network interruptions and forwarding once connectivity is restored.

OTA Configuration & Firmware Updates

Supports remote configuration, diagnostics, and firmware upgrades without physical access to the device.

Industrial-Grade Hardware

Built with reliable components, stable power architecture, and rugged enclosure for 24x7 operation.

Flexible Deployment & Mounting

Compact form factor with wall and ceiling mounting options for rapid indoor installations.

Suited Applications

Smart Buildings & Offices

Aggregates data from BLE sensors for indoor air quality, occupancy monitoring, energy analytics, and building management systems over Wi-Fi.

Retail & Shopping Centers

Supports BLE-based people counting, footfall analytics, proximity marketing, and in-store asset tracking.

Smart Campuses & Educational Institutions

Ideal for BLE-enabled attendance systems, indoor navigation, safety monitoring, and facility analytics across large campuses.

Warehouses & Logistics Facilities

Enables real-time tracking of BLE tags on pallets, racks, tools, and assets with centralized Wi-Fi backhaul.

Public Indoor Venues

Supports BLE-based monitoring and analytics in airports, metro stations, malls, and large indoor public spaces.

Hospitals & Healthcare Facilities

Enables reliable connectivity for BLE medical devices, asset tracking tags, patient monitoring wearables, and compliance sensors.

Hotels & Hospitality Spaces

Connects BLE room sensors, smart locks, occupancy sensors, and service devices to centralized hotel management platforms.

Industrial & Factory Interiors

Bridges BLE condition-monitoring sensors, equipment tags, and maintenance beacons to local servers and cloud platforms.

Data Centers & Server Rooms

Collects data from BLE temperature, humidity, and access sensors for continuous monitoring of critical indoor infrastructure.

Technical Specifications

General Specifications

Part Order Number	Setu WX1
Product Brand	Machman Technologies Pvt Ltd

Hardware Specifications

Processor Type	ARM Cortex Processor
Number of Cores	Dual Core
Clock Speed	960 MHz per Core
Memory Size	256 MB RAM

Memory Speed	Up to 1066 MHz
Internal Storage	4 GB eMMC
USB Port	Debug Interface (Bottom of Enclosure)
Connector Type	USB Type-C
Interface Protection (External*)	Surge Protection Enabled

BLE Specifications

Bluetooth Standard	Bluetooth Low Energy (BLE) 5.0 / 5.1
Operating Role	Central / Gateway
Supported BLE Devices	BLE Sensors, Beacons, Tags, Wearables, Edge Devices
Maximum BLE Connections	Multiple Concurrent BLE Devices (Configurable)
Data Handling	Periodic Scanning, Event-Based Collection, On-Demand Reads
Data Forwarding Protocols	MQTT, HTTP, HTTPS, TCP/IP
Local Buffering	Supported (Store-and-Forward during Network Outages)
Transmission Range (Indoor)	Up to 30–50 meters (Environment Dependent)
Security	Secure BLE Pairing, Encrypted Wi-Fi Backhaul

Backhual Ethernet

Number of Ports	x 1
Type	1 x 10/100M

Backhual WiFi

Bands	2.4 GHz ISM Band 5 GHz ISM Band
Standards	IEEE 802.11 b/g/n, 802.3, 802.3u, 802.11e-compatible bursting
Modulation Schemes	BPSK, QPSK, 16 QAM, 64 QAM
Security	WEP, TKIP, AES, WPA, WPA2
Antenna	Internal Antenna Integrated in Enclosure

Backhual Cellular Lte*

Cellular Bands	B1 / B3 / B5 / B7 / B8 / B20 / B28
Standards	3GPP LTE, IEEE 802.3, IEEE 802.3u, 802.11e-compatible bursting
Modulation Schemes	BPSK, QPSK, 16-QAM, 64-QAM
Security	WEP, TKIP, AES, WPA, WPA2
Antenna	Internal Antenna Integrated in Enclosure

Supported Cloud & Platforms

Cloud Connectivity	Public Cloud Platforms Private Cloud Deployments On-Premise Servers
Protocol Support	MQTT Brokers HTTP / HTTPS APIs TCP/IP Based Systems
IoT & Data Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud IoT (via MQTT / HTTP) Custom IoT Platforms
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines
Integration Method	REST APIs, Webhooks, MQTT Topics

Physical Characteristics

Power Supply	9-36 V PoE
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP35
Housing Material	Plastic
Color	Matte White
Weight	~140g
Mounting Options(Brackets)	Pole Mounting / Wall Mounting

Electrical Characteristics

Power Input	PoE IEEE 802.3af / 802.3at or 12–24 V DC
Power Consumption (Typical)	4.5 W (with Ethernet Backhaul)
Power Consumption (Peak)	Up to 7 W (with LTE & Wi-Fi Active)
Lightning / Surge Protection	Integrated surge protection on power & RF ports

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes (Nodes)
Firmware Protection	Secure boot and signed OTA updates

Setu WX1 Variants

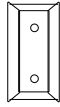
VARIANT NAME	POWER SOURCE	WIFI	ETHERNET	CELLULAR	IDEAL FOR
Setu WX1	12–24 V (Power)				Ideal for small indoor networks with stable Wi-Fi backhaul such as offices, retail stores and hospitality environments.
Setu WX2	12–24 V (Power)				Best for enterprise networks requiring reliable wired Ethernet connectivity and expanded indoor coverage in commercial buildings.
Setu WX3	12–24 V (Power)				Designed for mission-critical deployments where Ethernet, Wi-Fi and LTE failover ensure uninterrupted connectivity in industrial and multi-site facilities.

What's in the Box?

Make sure box contains following items before installations



Setu X1



**Mounting
Clip***



**User manual &
Warranty Card**



**IP
Extender***



**Mounting
Screws**

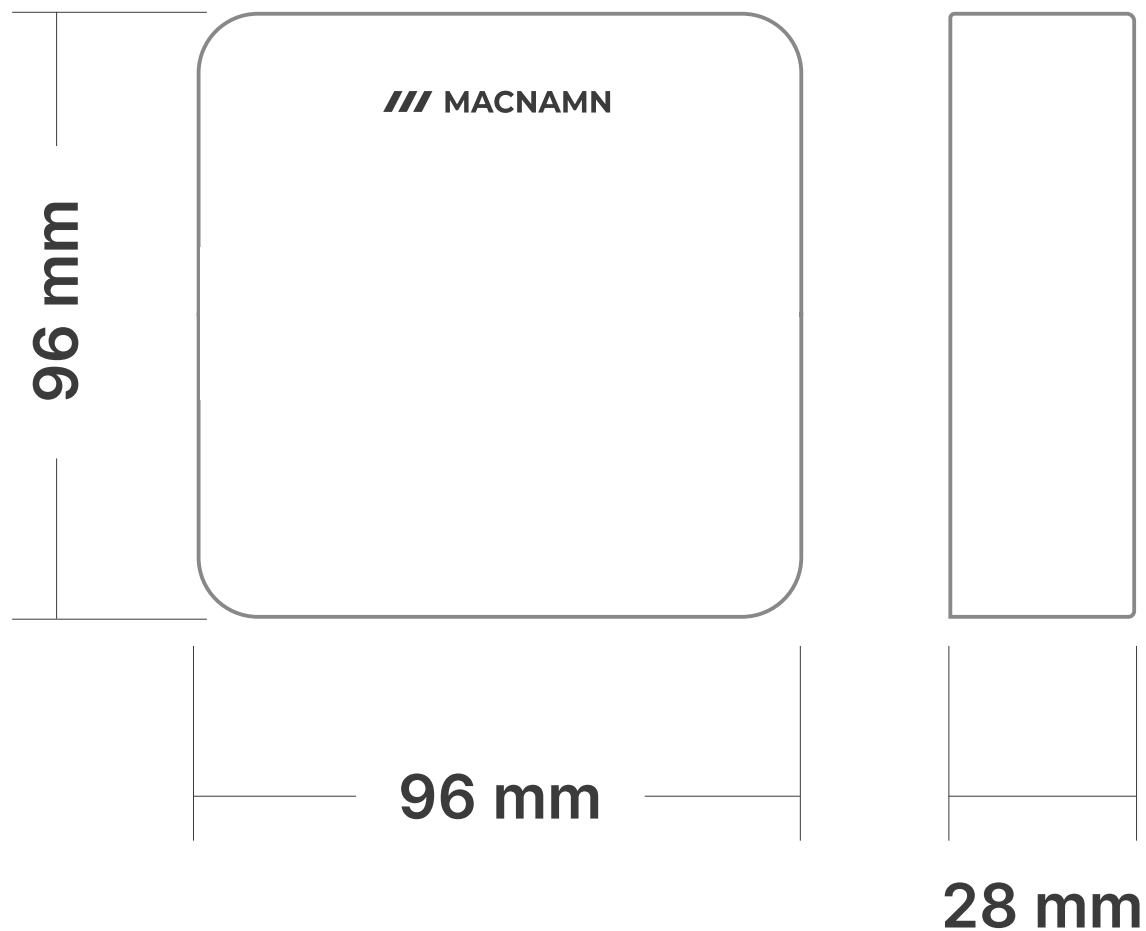


**Power
Adapters***

ⓘ Note

Accessories must be purchased separately & may not be included in the box.

Setu WX1 Dimensions



Frequently Asked Questions

What is the Setu WX1 used for?

Setu WX1 is a compact industrial BLE to Wi-Fi IoT gateway that bridges Bluetooth Low Energy sensors and tags to local servers and cloud platforms. Typical applications include Smart Buildings & Offices, Hospitals & Healthcare Facilities, Retail & Shopping Centers, Hotels & Hospitality Spaces, Smart Campuses & Educational Institutions and Industrial & Factory Interiors.

Which BLE devices does the Setu WX1 work with?

The Setu WX1 supports Bluetooth Low Energy (BLE) 5.0 / 5.1 and collects data from BLE Sensors, Beacons, Tags, Wearables, Edge Devices. It forwards data using MQTT, HTTP, HTTPS, TCP/IP. Indoor BLE range: Up to 30–50 meters (Environment Dependent).

Which backhaul options does the Setu WX1 have?

The Setu WX1 offers Ethernet, Wi-Fi and cellular LTE (optional) backhaul. Cellular bands: B1 / B3 / B5 / B7 / B8 / B20 / B28.

How is the Setu WX1 powered?

The Setu WX1 is powered from PoE IEEE 802.3af / 802.3at or 12–24 V DC. Typical power consumption is 4.5 W (with Ethernet Backhaul), with a peak of up to 7 W (with LTE & Wi-Fi Active).

What is the IP rating and operating temperature of the Setu WX1?

The Setu WX1 has an ingress protection rating of IP35 and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

Which cloud platforms does the Setu WX1 work with?

The Setu WX1 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud IoT (via MQTT / HTTP), Custom IoT Platforms. Integration methods: REST APIs, Webhooks, MQTT Topics.

Related Products and Resources

- Related datasheets: [Tag X3 BLE sensor tag and beacon](#), [BLE Tag X2 industrial BLE asset tracker](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)